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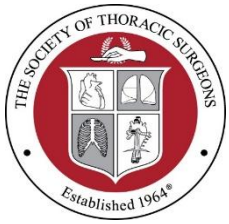


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**RE: Request for Information; Health Technology Ecosystem [CMS-0042-NC]**

On behalf of The Society of Thoracic Surgeons (STS), I write to provide comments on the Request for Information (RFI); Health Technology Ecosystem. Founded in 1964, STS is a not-for-profit organization representing more than 7,800 surgeons, researchers, and allied health care professionals worldwide who are dedicated to ensuring the best possible outcomes for surgeries of the heart, lungs, and esophagus, as well as other surgical procedures within the chest.

Below are some suggestions to improve the health technology ecosystem which could lead to better data integration, accuracy, and ultimately more informed clinical and operational decision-making.

**Improve Access to Federal Claims Data**

Clinician-led clinical data registries, such as the STS National Database, are crucial for enhancing quality improvement and conducting significant health care research. A clinician-led clinical data registry is a repository that is led by clinicians and is designed to collect detailed, standardized data on medical procedures, services, or therapies for diseases or conditions and provide feedback to participants. These registries are pivotal in gathering quality and clinical outcomes data to inform value-based care and must adhere to stringent quality and privacy standards to be effective.

However, the data collected by clinician-led clinical data registries are limited. For example, the STS National Database is restricted to a narrow 90-day or less post-treatment window. This constraint severely restricts our ability to monitor patient outcomes beyond this brief period, hindering the evaluation of long-term

effectiveness. Therefore, it is critically important for registries to supplement their data with claims data to assess the impact of interventions over extended period. In fact, Section 105(b) of the Medicare Access and CHIP Reauthorization Act of 2015 (MACRA) directs CMS to provide Medicare claims data to registries for linkage with clinical outcomes data for risk-adjusted, scientifical analyses and research to support quality improvement or patient safety.

If provided meaningful access to federal claims data, clinician-led clinical data registries would better track patient outcomes over time, better assess the safety and effectiveness of medical treatments, and provide the information to assess the cost-effectiveness of alternative therapies.

One of the key barriers to accessing federal claims data is the Virtual Research Data Center (VRDC). While the VRDC is intended to serve as a platform through which registries can access Medicare claims data for research, its use is often limited. The process is slow, complex, and prohibitively expensive. Data access through the VRDC is restricted to narrowly defined research projects, and approvals can take months—or even years—without any assurance of success. The high cost alone often deters registries from pursuing access. To improve access to claims data, we urge CMS to eliminate VRDC fees and collaborate with stakeholders to develop a more affordable and efficient pathway. Overcoming these challenges is essential to enabling clinician-led registries to continue driving improvements in clinical outcomes and healthcare quality.

**Standardize Unique Device Identifiers**

STS is supportive of recent efforts to expand the definition of unique device identifiers (UDI) for standardized data collection across. Standardized data collection of UDIs allows for better device tracking, increased patient and device safety, and higher quality data. STS currently maintains the STS National Database, the largest cardiothoracic surgical database in the world. The Database has four components, each focusing on a different area of cardiothoracic surgery—Adult Cardiac Surgery, Congenital Heart Surgery, General Thoracic Surgery, and Mechanical Circulatory Support. Currently, the Adult Cardiac Surgery Database (ACSD) contains more than 8 million cardiac surgery procedure records and has more than 3,800 participating physicians, including surgeons and anesthesiologists, representing more than 95% of all adult cardiac surgery hospitals and practices across the United States.

Information gathered from the database is used for improving the quality of adult cardiac surgery, facilitating the generation of new knowledge, informing patients and their families regarding the risks of surgery, and providing data to multiple national governmental and non-governmental agencies. Expanding the UDI field in certified EHR technology (CEHRT) would allow STS to more thoroughly report UDIs on the devices we collect for research and reporting purposes as requested by the Food and Drug Administration (FDA). Additionally, expanding the UDI field in CEHRT, would allow STS to expand UDI data capture on temporary mechanical circulatory assist devices, as well as better assist industry partners with post-approval studies as required by the FDA. As CMS quality programs require CEHRT, we believe they could be influential in attaining this update.

To ensure accuracy for improved data quality, STS strongly recommends that scanning UDIs into the electronic health records (EHR) be required. In our experience with the STS National Database, there is an increased burden on sites that manually enter complex UDIs and often lead to typographical errors. To account for this, STS only receives UDIs from sites that scan into the chart to avoid errors that render the UDI data unusable. As a result, the STS ACSD currently only captures approximately 30% of the heart valve device UDIs and 15% of ventricular assist device UDIs.

**Strengthen Imaging Interoperability Standards**

Another opportunity to improve data quality relies on strengthening the standardization and interoperability of medical imaging data through broader and more consistent adoption of the Digital Imaging and Communications in Medicine (DICOM) standard.

DICOM is the internationally recognized standard for the formatting, storage, and transmission of medical imaging and associated metadata.[1] It supports a wide array of data types, including patient demographics, imaging protocols, discrete measurements, and unique device identifiers (UDIs), all of which are critical for outcomes research, quality reporting, and regulatory monitoring. DICOM's structured format allows for integration of imaging data into EHRs.

Despite its utility, DICOM adoption remains incomplete and inconsistent, particularly in subspecialty imaging domains. Devices often claim DICOM compliance yet support only a limited or proprietary subset of its features.[2] For example, a device may conform to the storage of images, while omitting key metadata elements like UDIs or measurement annotations. This fragmented implementation undermines interoperability and limits the usefulness of imaging data for downstream applications.

As a result, systems that are technically "DICOM-compliant" may be incompatible in practice, preventing seamless data exchange between imaging devices, PACS, and EHRs.[3] These issues obstruct clinical workflows, hinder registry reporting, and delay innovation in data science applications. If CMS is looking to improve data quality, better DICOM standardization and utilization is required. CMS should work ASTP/ONC to integrate DICOM standards into the Certified Health IT program, requiring certified EHRs and PACS to support full DICOM functionality.

Thank you for the opportunity to provide these comments. Please contact Molly Peltzman, Associate Director of Health Policy, at mpeltzman@sts.org should you need additional information or clarification.

Sincerely,

Joseph F. Sabik III, MD
President

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[1] National Electrical Manufacturers Association (NEMA). **Digital Imaging and Communications in Medicine (DICOM) Standard**. https://www.dicomstandard.org

[2] Chiang MF, Campbell JP, et al. **The Case for Standards in Ophthalmic Imaging to Enable Artificial Intelligence**. *Trans Vis Sci Tech.* 2021;10(7):23. https://doi.org/10.1167/tvst.10.7.23

[3] Koff DA, Bak P. Implementing DICOM standards: a user's perspective. *J Digit Imaging*. 2012;25(5):635–641. https://doi.org/10.1007/s10278-012-9490-3